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Section 1

Competitive Positioning

This section maps PawSanctuary's design decisions against the two leading titles in the merge-puzzle subgenre: **Travel Town** (Magmatic Games, 10M+ downloads) and **Tasty Travels** (Century Games). The goal is to adopt what works, differentiate where PawSanctuary has a structural advantage, and avoid direct feature overlap that we cannot win.

What PawSanctuary Adopts from Travel Town

- **Nested chain mechanics:** Some items require merging products from two different chains (Habitat mechanic — see Section 7). Creates cross-family interdependency and genuine board pressure.
- **Order board urgency:** Timed rescue requests with visual escalation at low time remaining (timer bar already turns red; adding a pulse animation closes the gap with Travel Town's UX).
- **Manual spawner energy model:** Travel Town shifted away from auto-producers in 2025 toward energy-gated manual spawners. PawSanctuary's kibble-cost spawners are already well-positioned.
- **Seasonal map areas:** Limited-time regions with exclusive chains and cosmetic animals are a proven urgency and replayability driver. PawSanctuary's Seasonal Events system is the direct equivalent.

How PawSanctuary Differentiates from Travel Town

- **Named animal families with emotional attachment:** Travel Town uses generic objects (tools, furniture, building materials). PawSanctuary's 'Pup to Primordial' arc is a character journey. This is the game's core competitive moat.
- **Sub-object buff system:** Sub-objects granting Speed Burst, Spawner Refill, Map Supplies, and High-Tier Drop have no equivalent in Travel Town. A second-order differentiator that rewards engagement with the spawner economy.
- **Ambassador / milestone title arc:** Junior Rescuer → Animal Friend → Sanctuary Guardian → Legendary Rescuer creates a player identity layer Travel Town does not have. Titles earned from meaningful achievement drive social sharing.

What PawSanctuary Adopts from Tasty Travels

- **Global community events:** Tasty Travels' cooperative sanctuary challenges drive daily engagement beyond solo play. PawSanctuary's Community Rescue Events (Section 5) implement this.
- **Building-unlock streaks:** Completing N orders in a row without skipping unlocks bonus spawners or cosmetic upgrades. Worth integrating into PawSanctuary's AdoptionBoard flow as a retention layer.
- **Cooperative goals:** A shared weekly target (e.g., 'Rescue 10,000 animals this weekend') creates a reason to log in even when personal kibble is low.

PawSanctuary's Unique Angle

- **Living animal rescue theme:** Every item on the board has a name, a stage narrative, and emotional resonance. No competitor in this genre operates at this level of thematic coherence.
- **Family-specific passive powers (Superpowers):** 15 unique passive/active abilities unlock as families progress. No equivalent system exists in Travel Town or Tasty Travels.

- **Multi-family Habitat merges:** Combining two high-tier animals from different families to create a new item type is architecturally novel in the merge-2 genre.

Section 2

Phase 5: Map Area Expansion (9 Remaining Families)

Nine new SanctuaryArea entries are added to SanctuaryMap.swift, unlocking the remaining animal families. Build costs scale to create a clear sense of late-game investment. The Aquatics milestone path is retired per F.Q4 — Aquatics unlocks via its map area purchase.

Family	Species Key	Area Theme	Notes
Ursids (Bears)	.owl*	Bear Sanctuary	New area unlock
Aquatics (Fish)	.fish	Aquatic Habitat	Replaces 50-star milestone unlock (F.Q4)
Amphibians	.lizard*	Bog & Wetlands	New area unlock
Marsupials	.ferret*	Outback Refuge	New area unlock
Primates	.parrot*	Jungle Canopy	New area unlock
Equines (Horses)	.pony*	Open Pasture	New area unlock
Pachyderms	.hedgehog*	Savanna Grounds	New area unlock
Bovines (Cattle)	.guineaPig*	Farmland Haven	New area unlock
Cervids (Deer)	.fox*	Forest Clearing	New area unlock

* Species key assignments are placeholders pending AnimalSpecies enum extension.

New UpgradeBonus Types (Phase 5)

Bonus Field	Effect	Requires Code Change
subObjectDropRateBonus	Increases 20% base sub-object drop rate per activation	New field in UpgradeBonus struct
pityTimerReduction	Reduces spawn count required for Rare/Epic pity	New field in UpgradeBonus struct
powerUpDurationExtension	Extends active power-up duration (Speed Burst, etc.)	New field in UpgradeBonus struct

Build Cost Scaling

Current 6 areas cost tier-5 material combinations (hardcoded). Phase 5 areas will cost tier-5 and tier-6 materials in increasing quantities, following a logarithmic progression: each successive area costs roughly 1.4× the previous area's material total. Later areas require cross-type mixes (e.g., Wood tier 6 + Crystal tier 4) to create combinatorial depth in the build economy without requiring a material farming overhaul.

Implementation Notes

- Add 9 new SanctuaryArea enum cases to SanctuaryMap.swift (purely additive).
- Add 3 new fields to UpgradeBonus struct; update recalcActiveBonuses() with new handling.
- If Aquatics milestone path is retired, make placeRareSpeciesSpawnerForMilestone() a no-op or remove it; award a different reward at the 50-star milestone.
- Each new area needs its newFamilySpawner reward wired to the correct AnimalSpecies case.

Section 3

Phase 6: Superpower System

When a family first reaches **Era 3 (Stage 7)**, a unique passive or active ability unlocks for that family. Superpowers create long-term build diversity — they give experienced players a reason to develop all 15 families rather than specializing in two or three.

Family	Type	Name	Ability
Canines	Passive	Fetch!	Every 5th merge, a free Stage-1 Canine appears in a random empty cell
Felines	Active 90s	Nine Lives	Undo the last move — swap two pieces back to their prior positions
Rodents	Passive	Hoard	Rodent Spawner drops Stage-2 sub-objects instead of Stage-1
Avians	Passive	Scout	Preview shows the species of the next 2 incoming rescues
Bovines	Active 3min	Stampede	Instantly merge all same-family same-stage pairs on the board
Equines	Active 60s	Sprint	Doubles Kibble regen rate for 60 seconds
Ursids	Passive	Hibernate Bonus	If no merges for 5+ min, next rescue spawns at Stage 3
Cervids	Passive	Antler Drop	Every Cervid merge has a 25% chance to drop a bonus Stage-2 sub-object
Aquatics	Passive	Current	After any merge, sub-objects auto-slide to fill the nearest gap
Reptiles	Passive	Bask	Reptile Spawner charges regenerate at 2x the normal rate
Amphibians	Active 2min	Leap	Teleport any Amphibian to any empty board cell
Primates	Active 90s	Mimic	Spawns a free Stage-1 animal matching the last merge's family
Pachyderms	Passive	Memory	Quest and challenge progress for Pachyderm goals counts double
Lagomorphs	Passive	Multiply	Every 4th Lagomorph merge spawns two Stage-1 Lagomorphs
Marsupials	Active 3min	Pouch	Store up to 2 extra animals outside inventory for 30 seconds

Implementation Notes

- Add a new field on SanctuaryArea (or a separate superpowerState dict on GameState) to track whether a family's superpower has been unlocked.
- Passive effects: integrate into KibbleEngine (Sprint regen doubling), MergeBoardViewModel (Fetch!, Lagomorphs Multiply, Cervids Antler Drop), or BoardStateManager (Aquatics Current).
- Active effects: require a per-family cooldown timer and a button strip UI that expands dynamically as more active abilities are unlocked.
- The active superpower button strip is the primary engineering challenge — it must not clutter the board when empty and must scale cleanly as up to 6 active abilities become available.

Section 4

Seasonal Events System

Four seasonal events per year introduce time-limited content, exclusive currency, and cooperative urgency that drives daily active engagement beyond the core progression loop.

Event	Theme	Example Exclusive Chain	Timing
Spring Rescue	Baby animals, new arrivals	Bunny Bloom (4-stage)	March–April
Summer Safari	Exotic animals, open plains	Savanna Cub (4-stage)	June–July
Autumn Harvest	Woodland creatures, harvest	Forest Fox (4-stage)	September–October
Winter Warmth	Arctic/cozy animals	Festive Fox / Polar Cub	December

SeasonalEventManager

- Loads event config from a bundled JSON file (or remote fetch for live updates without an app release).
- Exposes a published activeEvent: SeasonalEvent? that drives tab visibility and event UI injection.
- Registers the seasonal chain in ContentRegistry only while the event is active.
- Manages event token accumulation and prize track threshold checking.

SeasonalEvent Model

Field	Type	Description
id	String	Unique event identifier (e.g., 'winter_2026')
startDate / endDate	Date	Event active window
chainID	ChainID	Temporary 4-tier chain registered during event
eventCurrencyName	String	'Event Tokens', 'Snowflakes', etc.
prizeThresholds	[Int: Reward]	5 reward tiers at token counts
orderBoardEnabled	Bool	Whether seasonal order board is active

Event Board

The Event Board lives in a separate tab alongside the main board and auto-dismisses when the event ends. Players keep all prizes claimed during the event window; unclaimed event tokens expire at event end. Board tab switching uses the same UI primitive as the InventoryScreen tab pattern.

Section 5

Social & Community Layer

Sanctuary Leaderboard

A weekly ranking of players by ambassador count using Game Center leaderboard infrastructure (partially in place via the existing authenticateGameCenter flow). Resets Sunday midnight. The leaderboard reinforces the Ambassador system as a prestige signal beyond solo play.

Community Rescue Events

Time-limited cooperative goals where each player contributes their ambassador completions during an event window. Example: 'Rescue 10,000 animals as a community this weekend.' Community milestone rewards are granted to all participants at 25%, 50%, 75%, and 100% of the goal. This mechanic drives log-ins even when a player's personal kibble is depleted.

Friend Gifting

Send one free kibble gift per day to each Game Center friend. Receiving a gift adds 5 kibble (intentionally below the regen cap threshold to ensure it never fully refills). Uses the existing InviteSystem infrastructure for friend discovery. The 5-kibble amount is generous enough to feel meaningful without replacing the need for IAP.

Feature	Implementation Path	Target
Weekly Leaderboard	Extend GameCenterCoordinator; Game Center leaderboard	Q4 2026
Community Events	New CommunityEvent model; server-side goal accumulation	Q1 2027
Friend Gifting	Extend InviteSystem; Game Center friend API + gift state	Q4 2026

Section 6

Three-Tab Inventory (GDD v3.0)

The three-tab inventory replaces the current visual-tab / shared-slot-pool implementation with three genuinely separate slot pools. The three-tab constraint creates natural board pressure that drives shop conversions without ever feeling punitive.

Tab	Contents	Default Slots	Unlock Cost	Key Feature
Tab 1 — Sanctuary Animals	Animal board items (merge staging)	10 slots	Free (always open)	Primary merge staging area; unlockable rows
Tab 2 — Spawners	Sub-object chain items in transit	6 slots	10 Dog Tags	Dedicated sub-object storage
Tab 3 — Building Supplies	Build materials (wood, stone, crystals)	Unlimited	25 Dog Tags*	Auto-merge; tier-count display; no slot cap

* Requires Tab 2 to be unlocked first.

Tab 3 — Building Supplies Detail

- Materials displayed as tier counts with merge progress indicator (e.g., 'Wood Tier 3: 7') — not individual item slots.
- Two materials of the same tier auto-merge into the next tier when both are present.
- No slot cap — the tab accumulates indefinitely with display paging for large collections.
- Eliminates the board-clogging problem of building materials appearing as individual items.

Implementation Notes

- InventoryStore needs three genuinely separate [BoardItem?] arrays (not one shared pool with visual tabs).
- addItem(_) routing: .animal → Tab 1, .subObject → Tab 2, .material → Tab 3 (count-based, not slot-based).
- Tab 3 auto-merge logic runs on every addItem call for materials; no user gesture required.
- Migration: existing saves with shared slot pool must be split into the two explicit pools on first launch after update.

Section 7

Nested Chain / Habitat Mechanic

Three 'Habitat' chains create late-game goals that extend meaningfully beyond Ambassador progression. Each requires combining high-tier animals from two different families — the only cross-family merge mechanic in the game. This draws directly from Travel Town's nested chain design while remaining thematically coherent with PawSanctuary's rescue narrative.

Habitat	Input A	Input B	Output	Reward on Completion
Forest Habitat	Cervid tier 5+	Ursid tier 5+	Forest Scene Tier 1–4	+10% Kibble regen (both families) + Forest Clearing m
Ocean Habitat	Aquatics tier 5+	Amphibian tier 5+	Ocean Scene Tier 1–4	+10% Kibble regen (both families) + Tide Pool map de
Savanna Habitat	Bovine tier 5+	Equine tier 5+	Savanna Scene Tier 1–4	+10% Kibble regen (both families) + Watering Hole m

Strategic Value

- **Late-game goal extension:** A player who has completed all 15 family ambassador chains still has 3 Habitat chains to pursue.
- **Cross-family strategy:** Completing a Habitat requires investing in two families simultaneously, creating richer board management decisions.
- **Permanent economic bonus:** The +10% Kibble regen bonus makes Habitat completion a meaningful upgrade, not purely cosmetic.
- **Map decoration signal:** Visible decorations on the sanctuary map signal mastery to the player and (eventually) to friends.

Implementation Notes

Habitat chains use the same BoardItem(chainID, tier) model as all other chains, with chainIDs 'habitat.forest', 'habitat.ocean', and 'habitat.savanna'. The only new mechanic is the cross-family merge input: attemptMergeOrMove needs a new branch that detects two items from different specified families when both meet the minimum tier threshold (tier 5+) and produces the corresponding Habitat Tier 1 item. This branch fires before the standard same-chainID merge check.

Section 8

Monetization Roadmap

Current Monetization

- Sanctuary Pass subscription (\$4.99/month): +30 Kibble/day, free daily rescue request skip, free daily Spawner Refill.
- IAP Kibble bags: \$0.99 / \$1.99 / \$4.99 for 60 / 180 / 600 Kibble.
- IAP Dog Tag packs: \$0.99 / \$2.99 / \$6.99 for 15 / 60 / 175 Dog Tags.
- Sanctuary Starter Pack: 100 Kibble + 20 Dog Tags for \$1.99.
- Rewarded ads (planned): +5 Kibble, max 3×/day; shown only when Kibble is at 0.

Horizon	Feature	Revenue Mechanism
Now	Sanctuary Pass + existing IAP	Recurring subscription + one-time Kibble/Tags
Near-term	Seasonal event currency IAP bundles	Event token packs; limited-time starter packs per season
Near-term	Rewarded ads	CPM revenue; max 3/day to avoid fatigue
Medium-term	Habitat cosmetic variants	Premium animal skins per Habitat tier (cosmetic only)
Medium-term	Seasonal battle pass variant	Event-specific pass with prize track at \$2.99–\$4.99
Long-term	Guild / cooperative perks	Optional premium guild features for organized groups
Long-term	Animal encyclopedia IAP	Unlock full encyclopedia vs. only discovered entries

Monetization Principles

- No paywalled story content or hard progression gates.
- No 'pay to win' — Dog Tags and Kibble never give board advantages over other players.
- Core gameplay must always be completable free-to-play; spending accelerates, not enables.
- Per-unit Kibble cost is 30–40% cheaper than Travel Town equivalent at every price point.
- Never show a purchase prompt more than once per session.

Section 9

Art & Asset Pipeline

Current State

The current build uses SF Symbols as placeholders throughout the board, inventory, quest UI, and spawner tiles. All tint colors and family-specific symbols are defined in `AnimalSpecies.swift` (`sfSymbol`, `tintColor` fields). No custom art assets are in the project.

Required Art Assets

Asset Category	Count	Priority
Animal sprites (15 families x 15 tiers)	225 unique sprites	Critical — visible to all players
Sub-object sprites (15 families x 4 tiers)	60 unique sprites	High — visible in active play
Habitat tier sprites (3 habitats x 4 tiers)	12 unique sprites	Medium — Phase 7+ feature
Building sprites (map area icons)	15 sprites	Medium — visible on map screen
UI icons (power-up effects, currency, tabs)	~30 icons	High — core UI clarity
Spawner tile art (15 families)	15 unique tiles	High — visible in active play
Background music tracks (per area)	6–9 tracks	Medium — post-launch polish
Sound effects (SoundManager — 6 events)	6 audio files	High — merge feel

Recommended Delivery Order

- **Phase A — Tiers 1–5 for all 15 families** (75 sprites): these are visible to every player from session 1. Highest coverage-per-sprite ROI.
- **Phase B — Tiers 6–10 for all families** (75 sprites): mid-game content seen by retained players. Deliver within 60 days of launch.
- **Phase C — Tiers 11–15 for all families** (75 sprites): late-game Legendary art seen by a smaller but highly engaged segment. Can be delivered post-launch.
- **Phase D — Sub-objects, Habitats, UI polish** (60+ assets): non-blocking for launch but required for full v2.0 feature set.

Integration Notes

The content-registry pattern means adding art assets requires only replacing the `sfSymbol` string with an asset catalog reference in `AnimalSpecies.swift` — no code changes. The registry lookup falls back to SF Symbols if a custom asset is not yet available, making partial art delivery safe.

Section 10

Implementation Priority Matrix

The matrix below orders all planned features by urgency, complexity, and player impact. Complexity is rated Low / Medium / High / Very High. Player Impact is rated on the dimension with the highest leverage (stability, content, retention, or virality).

Priority	Feature	Phase	Complexity	Player Impact
1	BUG-01/02/03 fixes (notification delegate, persist now claim, timer Rollback Rollback)			High — stability
2	Phase 5 Map Expansion (9 remaining families)	Next	Medium	High — new content
3	Three-Tab Inventory UI (separate slot pools)	Next	Medium	Medium — UX clarity
4	Seasonal Events System (SeasonalEventManager, Q3 2026 Board)	Q3 2026	High	Very High — retention
5	Phase 6 Superpowers (15 abilities, active UI strip)	Q3 2026	Medium	High — engagement
6	Social Leaderboard (Game Center, weekly reset)	Q4 2026	Low	High — virality
7	Habitat / Nested Chain Mechanic (3 biome chains)	Q4 2026	High	High — late-game depth
8	Community Rescue Events (cooperative goals)	Q1 2027	High	Very High — retention
9	Art Asset Pipeline (225 animal sprites + 60 sub-object animations sprites)		Very High	Very High — core experience

Decision Log (Locked)

Ref	Question	Decision
F.Q1	9-tier vs. 15-tier chains	Replace — new 15-tier stage names supersede old 9-tier names
F.Q2	Kibble economy rebalancing	Regen cap 100 → 150 at player level 10
F.Q3	Sell values for tiers 9–14	1,000 / 2,000 / 5,000 / 10,000 / 25,000 / 100,000 coins
F.Q4	Aquatics unlock path	Map area replaces the 50-star milestone unlock
F.Q5	Pity timer thresholds	30 spawns → Rare guarantee; 60 spawns → Epic guarantee
F.Q6	Inventory expansion	Building Supplies tab is limitless; auto-merge on receipt
F.Q7	Superpower system scope	Deferred to Phase 6
F.Q8	Sub-object base drop rate	20% base rate per spawner tap; 60/25/10/5 split within that 20%
F.Q9	New upgrade bonus types	Sub-object drop rate, pity timer reduction, power-up duration extension
F.Q10	Board size	9x7 = 63 cells confirmed; GDD 6x5 references are superseded

End of PawSanctuary Feature Expansion Plan & Roadmap v2.0 — June 2026. All decisions in the Decision Log are locked for implementation unless re-opened with new evidence.